

Movie Success Factors in the Film Industry (2000-2024): A CRISP-DM Analysis of Budget, Revenue, Genre, and Language

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Introduction

Context

The global movie industry earns quite a significant amount of money through movies being watched at theaters or streamed online. It is therefore important for us to know what makes a movie successful. This report will make use of TMDb movies data between the years 2000 and 2024 to analyze factors influencing a movie's success using the CRISP-DM framework in two cycles (Bemthuis et al., 2025).

Research Objectives

First Cycle

- To identify overall patterns in movie release volume, ratings, and revenue between 2000 and 2024.
- To examine the relationship between a movie's budget and the revenue it generates
- To identify which movies achieved the highest revenue and the highest ratings, and to understand which features are most associated with movie success

Second Cycle

- To examine how a movie's genre relates to its ratings and commercial success
- To examine how a movie's original language relates to its ratings and commercial success

First CRISP-DM Cycle

Business Understanding

Rising production costs and tighter budgets have made financial sustainability an ongoing challenge for the film industry. Understanding how budget relates to a film's eventual success is therefore a key business question. This cycle focuses on that business context from the perspective of a specific stakeholder, breaking the overall question down into the sub-questions addressed below.

Stakeholder

This analysis is framed for a **streaming platform content-acquisition team**, responsible for deciding which types of films to license, promote, or commission. This stakeholder needs to understand which factors reliably predict commercial success (revenue) and critical/audience reception (rating), so that acquisition and production-partnership decisions can be prioritized accordingly.

Success Criteria

This analysis would be deemed a success if the following three factors were achieved: (1) identification of consistent patterns in the relationship between budget, genre, release timing, and both rating and revenue; (2) determining whether high ratings consistently correlate with high revenue or whether the two are relatively independent of each other; and (3) practical advice based on the data analysis for the stakeholder.

Research Questions

- What are the key patterns of the success factors of movies in the dataset used?
- How is the movie budget connected to the movie revenue in the film industry?
- What type of changes have the movies experienced over time in terms of release, revenue and rating patterns?
- Which movies have received the highest ratings and have generated the highest revenue?

Data Understanding

The dataset comprises 25 years' worth of yearly CSV files (from 2000-2024), containing the same set of features like id, title, genre, release year, budget, revenue, and votes. The datasets have been combined row-wise into one single file. These features have been selected since they are relevant to the requirements of the stakeholder: budget, revenue, profit, votes, genre, and release year.

Number of rows: 670208

Number of columns: 25

Table 1: Sample of Raw Merged Data

id	title	vote_average	vote_count	release_date	revenue	budget	runtime	original_language
98	Gladiator	8.209	16962	2000-05-04	465361176	1.03e+08	155	en
77	Memento	8.188	13723	2000-10-11	39723096	9.00e+06	113	en
8358	Cast Away	7.662	10531	2000-12-22	429632142	9.00e+07	143	en
36657	X-Men	6.996	10530	2000-07-13	296339527	7.50e+07	104	en
1359	American Psycho	7.400	9797	2000-04-13	34266564	7.00e+06	102	en
641	Requiem for a Dream	8.027	9165	2000-10-06	7390108	4.50e+06	102	en

The new dataset comprise 670,208 rows and 25 columns. The high number of observations over a 25-year period makes it possible to look for consistent patterns rather than just exceptions.

Data Preparation

Column names were standardised and columns were cast into proper data types. Duplicates based on movie id were deleted and now there are 670,072 number of rows in the movies data frame. Variables related to profit, roi, revenue-success, and rating-group were created and movies data frame was subsetting to only include the years 2000 to 2024. Inconsistent entries were replaced by NA instead of being deleted.

Table 2: Missing Values by Column

Variable	Missing_Count
roi	627849
homepage	572950
tagline	567864
keywords	468421
backdrop_path	450480
production_companies	344332
imdb_id	343030
production_countries	273520
spoken_languages	252551
genres	239355
runtime	147612
poster_path	143665
overview	110744
title	4
original_title	4
revenue	1
profit	1
source_file_id	0
id	0
vote_average	0
vote_count	0
status	0
release_date	0
adult	0
budget	0
original_language	0
popularity	0
release_year	0
release_month	0
release_day	0
revenue_success	0
rating_group	0

Modeling

Global Trends

After merging and cleaning the dataset, a simple statistical description offers a preliminary overview of the data to be analyzed further. The mean rating for 670072 movies produced in the period from 2000 to 2024 is 2.050817, while the mean revenue is 894739.4; the average budget equals 351558.2 and the mean duration is 68.77461.

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##
## ===== BASIC SUMMARY =====

## Total movies: 670072

## Year range: 2000 to 2024

## Average rating: 2.050817

## Average revenue: 894739.4

## Average budget: 351558.2

## Average runtime: 68.77461
```

To identify broad patterns before narrowing the focus in Cycle 2, the top-performing movies by revenue and rating were tabulated below, alongside yearly release counts and average revenue, with the corresponding trends examined in the plots that follow.

Table 3: Top 10 Revenue Generating Movies

title	release_year	revenue	budget	profit	vote_average	popularity
Adventures in Bora Bora	2023	3000000000	8.00e+08	2200000000	0.000	0.000
TikTok Rizz Party	2024	3000000000	2.50e+08	2750000000	10.000	0.000
Avatar	2009	2923706026	2.37e+08	2686706026	7.573	79.932
Avengers: Endgame	2019	2800000000	3.56e+08	2444000000	8.263	91.756
Avatar: The Way of Water	2022	2320250281	4.60e+08	1860250281	7.654	241.285
Star Wars: The Force Awakens	2015	2068223624	2.45e+08	1823223624	7.293	66.772
Avengers: Infinity War	2018	2052415039	3.00e+08	1752415039	8.255	154.340
Best Of Joy	2024	2000000000	1.00e+03	1999999000	0.000	0.000
Spider-Man: No Way Home	2021	1921847111	2.00e+08	1721847111	7.990	186.065
Jurassic World	2015	1671537444	1.50e+08	1521537444	6.682	54.089

Table 4: Top 10 Highest Rated Movies

title	release_year	vote_average	vote_count	revenue	popularity
Prison High Pressure	2019	9.200	210	0	297.563
BTS World Tour: Love Yourself - Japan Edition	2019	9.172	306	0	4.892
Break the Silence: The Movie	2020	9.154	175	8954945	7.198
BTS Permission to Dance On Stage - Seoul: Live Viewing	2022	9.018	113	32600000	6.877
Franco Escamilla: por la anécdota	2018	9.000	118	0	7.013
BTS: Permission to Dance on Stage - LA	2022	8.993	148	0	11.480
The Three Deaths of Marisela Escobedo	2020	8.904	215	0	8.083
Metallica & San Francisco Symphony: S&M2	2019	8.759	110	0	6.698
Chivas: The Movie	2018	8.655	113	0	5.148

title	release_year	vote_average	vote_count	revenue	popularity
BLACKPINK: Arena Tour 2018 ‘Special Final in Kyocera Dome Osaka’	2019	8.652	141	0	4.811

Table 5: Count of Movies Per Year

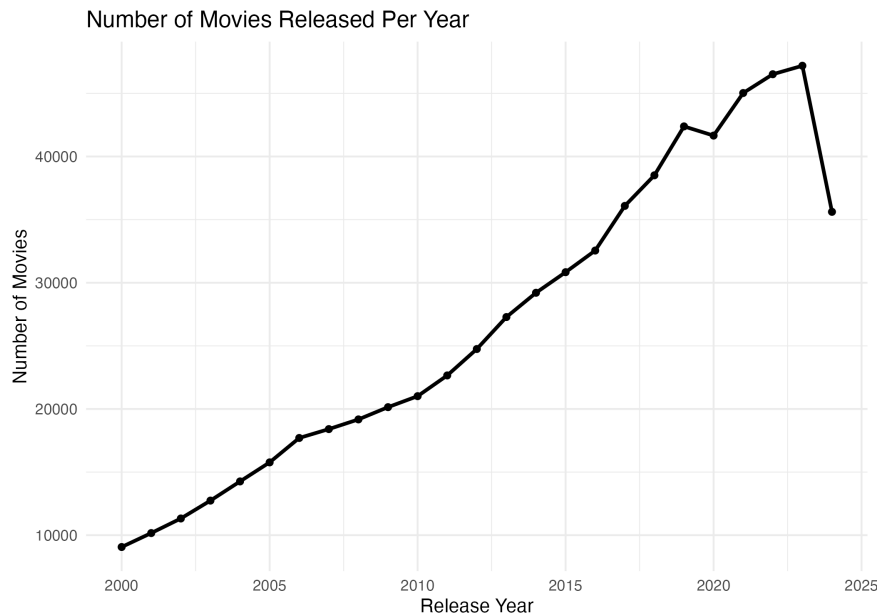
release_year	n
2000	9066
2001	10171
2002	11326
2003	12743
2004	14262
2005	15771
2006	17704
2007	18407
2008	19175
2009	20145
2010	21016
2011	22658
2012	24751
2013	27283
2014	29207
2015	30837
2016	32551
2017	36088
2018	38511
2019	42386
2020	41658
2021	45030
2022	46518
2023	47190
2024	35618

Table 6: Average Revenue Per Year

release_year	avg_revenue	total_movies
2000	1421973.1	9066
2001	1537335.3	10171
2002	1575080.9	11326
2003	1524395.7	12743
2004	1350951.7	14262
2005	1179745.8	15771
2006	1174869.8	17704
2007	1233752.5	18407
2008	1256332.4	19175
2009	1306485.2	20145
2010	1209525.3	21016
2011	1273173.2	22658
2012	1153215.4	24751
2013	1096160.9	27283

release_year	avg_revenue	total_movies
2014	1041021.4	29207
2015	1020166.7	30837
2016	1043425.5	32551
2017	943972.4	36088
2018	856827.6	38511
2019	759689.7	42386
2020	215126.1	41658
2021	378676.5	45030
2022	549097.4	46518
2023	574136.8	47190
2024	435081.4	35618

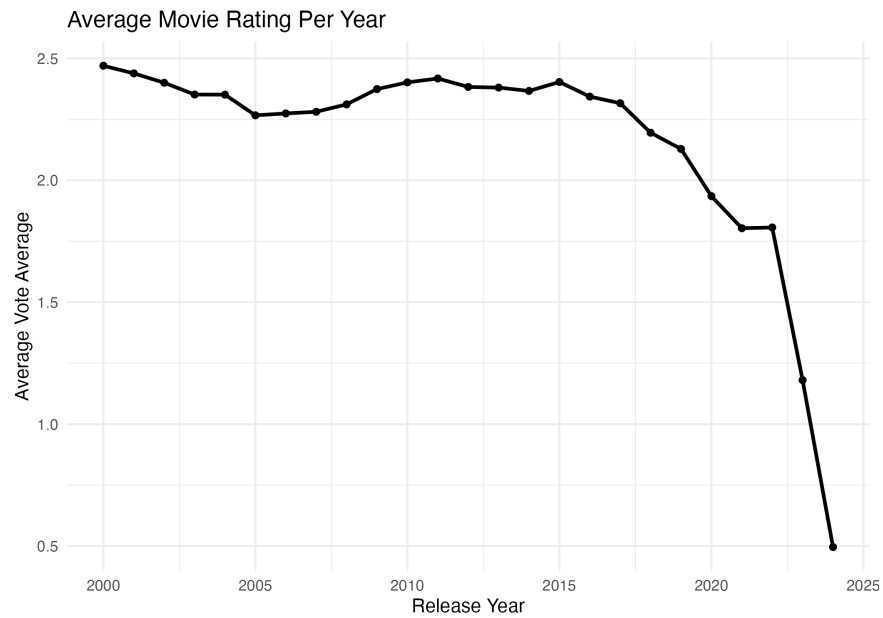
The number of movies produced every year increased gradually from 9,066 in the year 2000 to reach its highest point of 47,190 in the year 2023, followed by a decrease to 35,618 movies in 2024. A slight decrease can be observed in the year 2020 (41,658 movies compared to 46,518 in 2019). This might be due to a decline in the production and release of movies in the year 2020 due to the pandemic outbreak of COVID-19, as it recovered by the year 2021-2023. The steep fall after 2023 is much more than the fall observed in 2020.

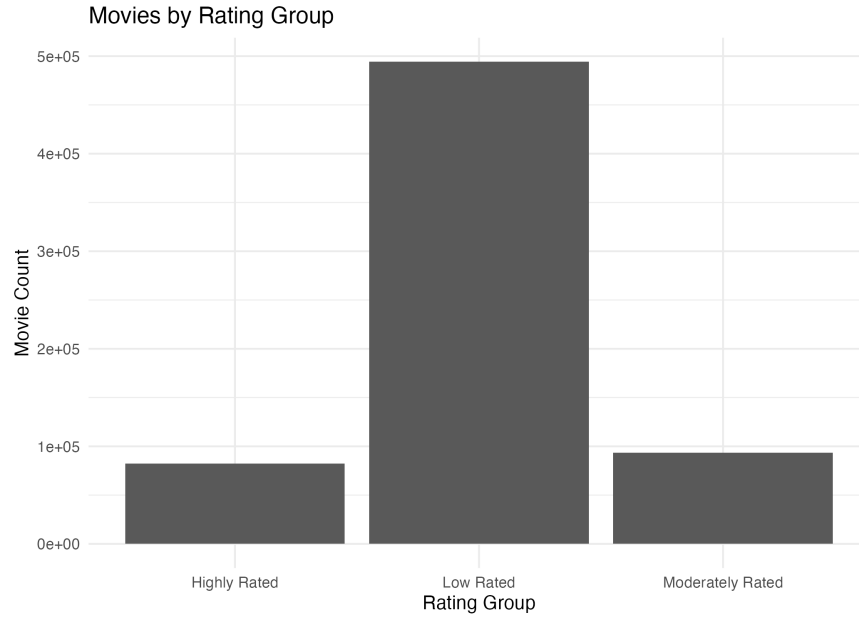


Average movie rating has dropped from 2.47 in the year 2000 to 0.50 in the year 2024, with the largest drop recorded after the year 2020 (from 2.15 in 2020 to 1.18 in 2023, and further down to 0.50 in 2024). Nevertheless, it would be better to interpret the above finding cautiously and not take it as direct evidence of declining movie quality. According to the Movies by Rating Group graph presented below, most of the movies in the dataset belong to the Low Rated group, and the TMDb's bulk data contains numerous low engagement or incomplete entries – some of which definitely indicate data quality problems rather than the release of new movies (see Study Limitations). Thus, the above mentioned decline in average ratings may indicate a growing share of these low-quality data among the more recent data.

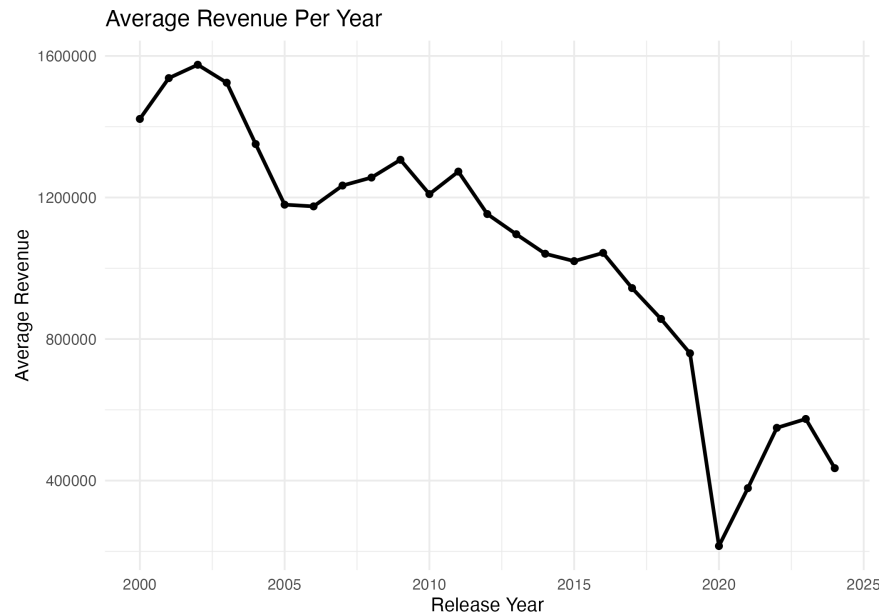
Table 7: Average Ratings Per Year

release_year	avg_rating	n_movies
2000	2.4696467	9066
2001	2.4384116	10171
2002	2.3999282	11326
2003	2.3516138	12743
2004	2.3511642	14262
2005	2.2663757	15771
2006	2.2740765	17704
2007	2.2808919	18407
2008	2.3112832	19175
2009	2.3740385	20145
2010	2.4015279	21016
2011	2.4173915	22658
2012	2.3827337	24751
2013	2.3801407	27283
2014	2.3662825	29207
2015	2.4030771	30837
2016	2.3432501	32551
2017	2.3160067	36088
2018	2.1949464	38511
2019	2.1285562	42386
2020	1.9348973	41658
2021	1.8033391	45030
2022	1.8064311	46518
2023	1.1804539	47190
2024	0.4960628	35618



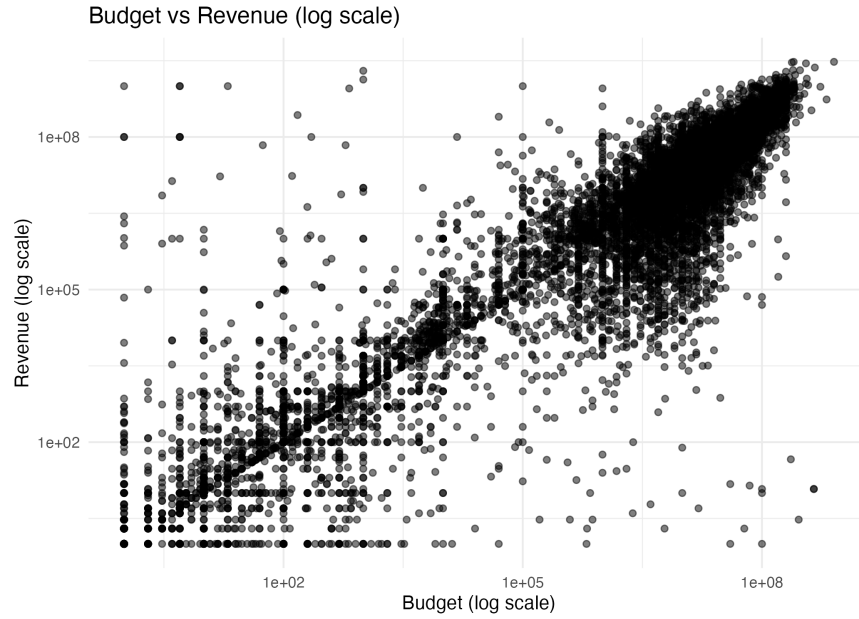


In line with the above trend, low-rated films greatly outweigh films that are of medium and high ratings, collectively making up about three-quarters of the entire dataset. This provides additional evidence that the decrease in the average rating with time is an indication of an increasing proportion of low-engagement and low-quality films.

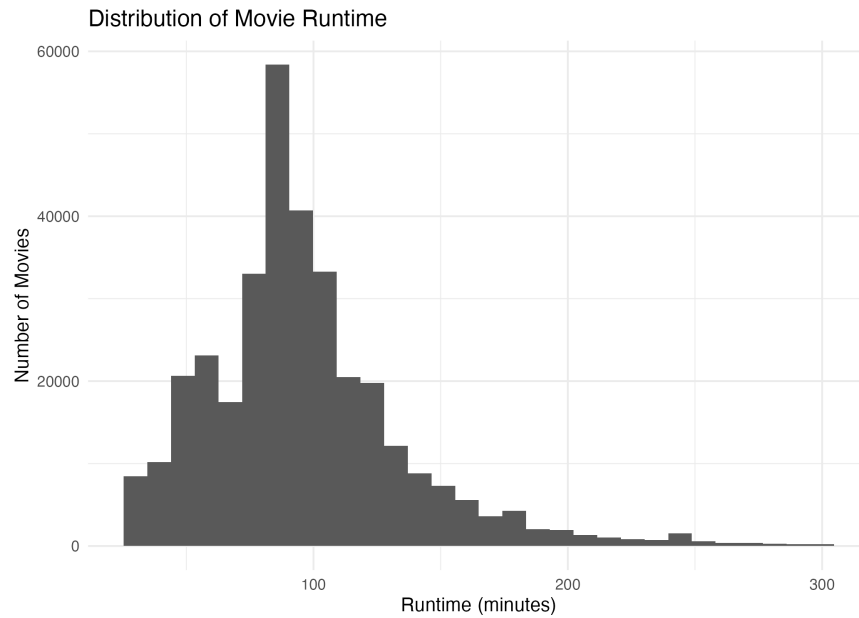


The average annual revenues went up from 1,421,973 in 2000 to 1,575,081 in 2002 and then fell steadily to 759,690 in 2019. There was a sudden fall in 2020 to 215,126, which can be attributed to the pandemic-driven closure of movie theatres, after which there was a slight bounce back in 2023 to 574,137 and then another fall to 435,081 in 2024.

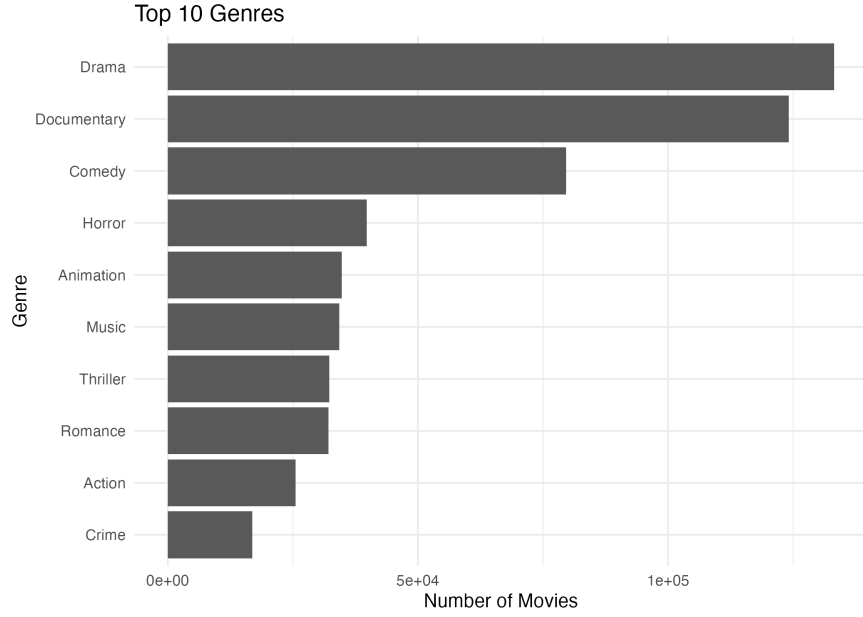
On the whole, the long-run trend throughout the entire period depicts falling average revenue, in line with the trend witnessed in the ratings presented above.



Drama, Documentary, and Comedy are the three most common genres by movie count, together accounting for a substantial share of all releases in the dataset. This indicates that recent film production continues to concentrate heavily on these established genres. Whether this high production volume translates into stronger commercial or critical performance is examined further in Cycle 2.



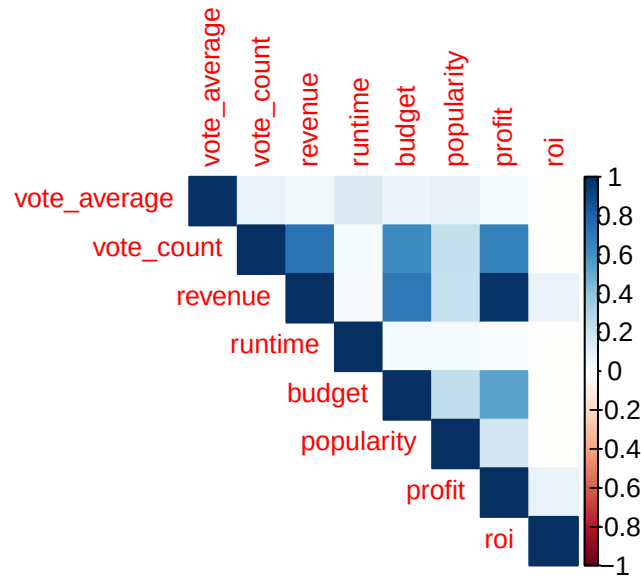
Most films have their duration distributed around a region between approximately 80 and 120 minutes, reflecting feature films' durations. Some films are much longer than this interval and constitute the tail of longer durations.



The correlation between revenue and profit is extremely high (0.97), which is understandable because profit is calculated from revenue. The correlation between the number of votes and revenue is also quite high (0.73), as well as the number of votes and the budget of the movie (0.64). However, this refers to the number of people who saw or voted for the movie, and not to how good the movie was because the correlation between the average value of the rating (`vote_average`), on the one hand, and revenue (0.06), budget (0.08) and profit (0.05), on the other hand, is extremely low. This means that the success and popularity of the movie are largely independent of its reception: a high rating does not necessarily lead to a high revenue and vice versa. Popularity is positively correlated with revenue (0.22) and the number of votes (0.23).

Table 8: Correlation Matrix

	vote_average	vote_count	revenue	runtime	budget	popularity	profit	roi
vote_average	1.0000000	0.0958849	0.0596094	0.1538229	0.0756965	0.1006427	0.0469618	-
vote_count	0.0958849	1.0000000	0.7300981	0.0456980	0.6356562	0.2327055	0.6729876	-
revenue	0.0596094	0.7300981	1.0000000	0.0346697	0.7195390	0.2229483	0.9724761	-
runtime	0.1538229	0.0456980	0.0346697	1.0000000	0.0498787	0.0416310	0.0256571	-
budget	0.0756965	0.6356562	0.7195390	0.0498787	1.0000000	0.2459807	0.5379253	-
popularity	0.1006427	0.2327055	0.2229483	0.0416310	0.2459807	1.0000000	0.1881045	-
profit	0.0469618	0.6729876	0.9724761	0.0256571	0.5379253	0.1881045	1.0000000	-
roi	-0.0071292	-	0.0817757	-	-	-	0.0999519	1.0000000
		0.0016258		0.0087336	0.0019635	0.0009876		



Evaluation - Cycle 1

Key Findings

- The number of releases increased from 9,066 (2000) to 47,190 (2023), but then decreased to 35,618 (2024).
- The average rating went down from 2.47 to 0.50 due to an increase in low-quality films.
- The average revenues have dropped throughout the years, especially since 2019.
- The budget and revenue show high correlation (0.72).
- The vote count is correlated with both revenue and budget, while the rating is not.

Link to Success Criteria

This data is directly linked to the success criteria where the budget and revenue relationship was determined to be (0.72 correlation), while the ratings-revenue relationship was confirmed as being negligible, thus making it independent from each other. This is the first step towards acquisition decisions, but more detailed information will be provided in Cycle 2.

Limitations

- There is low correlation among the vast majority of individual factors (except for budget and revenue) and the success of the movies, meaning that there are fewer conclusions that can be made on the basis of Cycle 1.
- There is an abundance of missing data for several columns (for example, homepage, tagline, keywords, production company details); however, the columns with the core variables utilized in this cycle have virtually zero missing data.

Questions for Cycle 2

- Which genre of movies makes the most money and has high ratings?
- Is there any specific effect of genre and language on movies' success, apart from the budget itself?

Second CRISP-DM Cycle

Refined Business Understanding

In Cycle 1, budget was found to be an important determinant of revenue generation; however, ratings were not found to be so. In this cycle, two additional variables that emerge from the findings in Cycle 1 and stakeholders' interest have been studied: genre and language.

Research Questions

- What movie genre is known to have high average revenue and ratings?
- How does the original language of a movie affect its revenue and ratings?

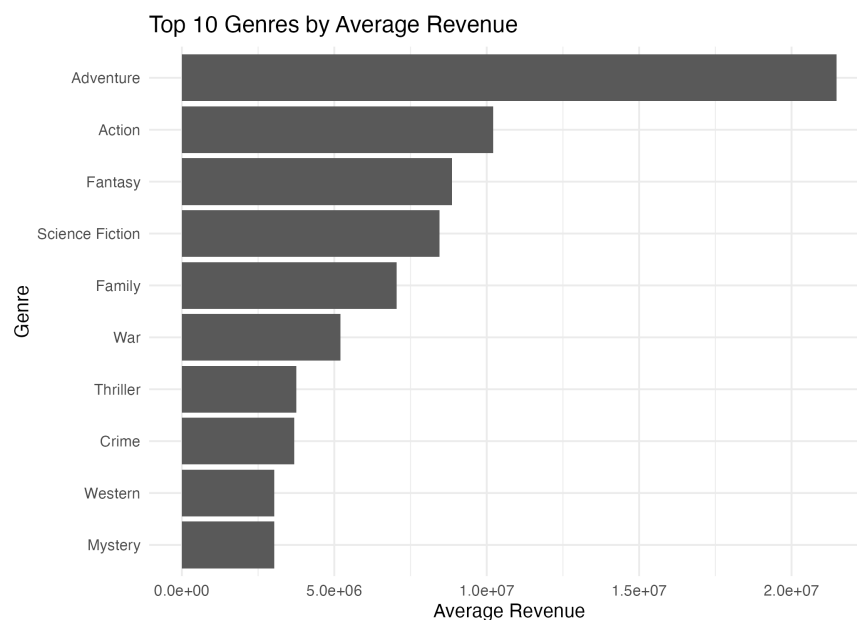
Enhanced Data Preparation

The dataset was limited to include id, title, genres, original_language, vote_average, and revenue. Missing data in the genre was dropped. As a movie can belong to many genres, each movie-genre combination was expanded to one line per genre in order for it to make its own contribution to the calculation of the average per genre separately.

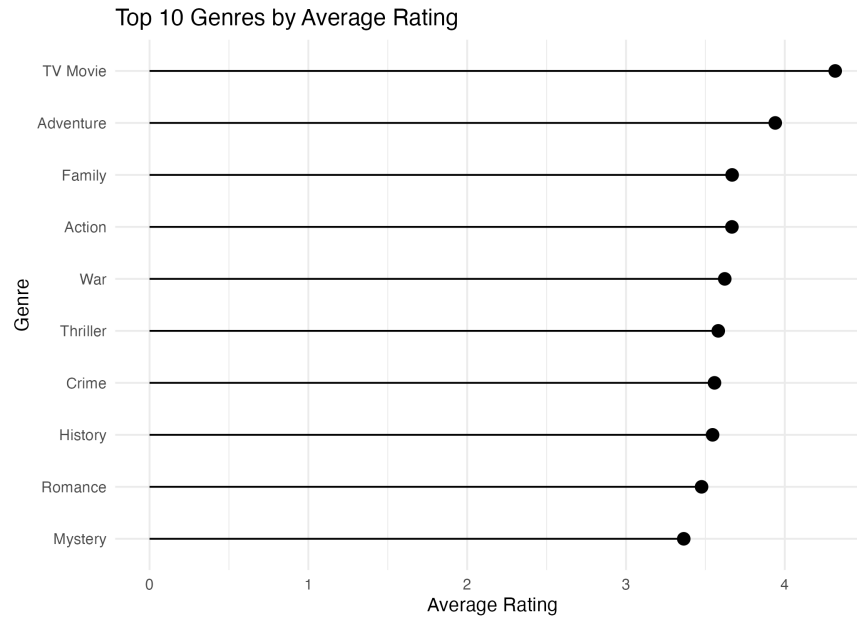
Advanced Modeling and Final Evaluation

Influence of Genre on Movie Success

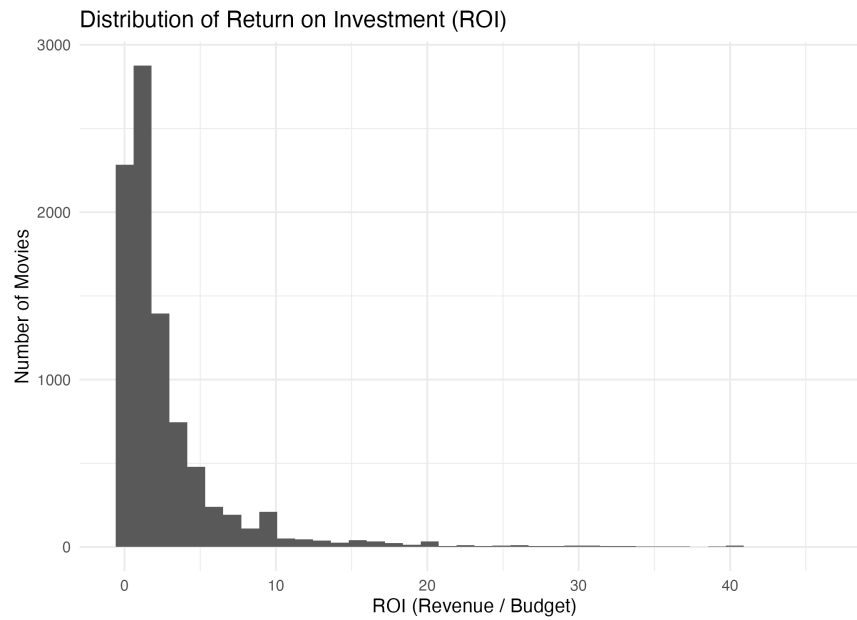
Adventure performs on both average earnings and ranking (second behind the small TV Movie subcategory). Drama has the largest number of movies, but is not a leader on either performance metric, indicating a possible overabundance compared to its value – meaning that Adventure-type films could yield better results..



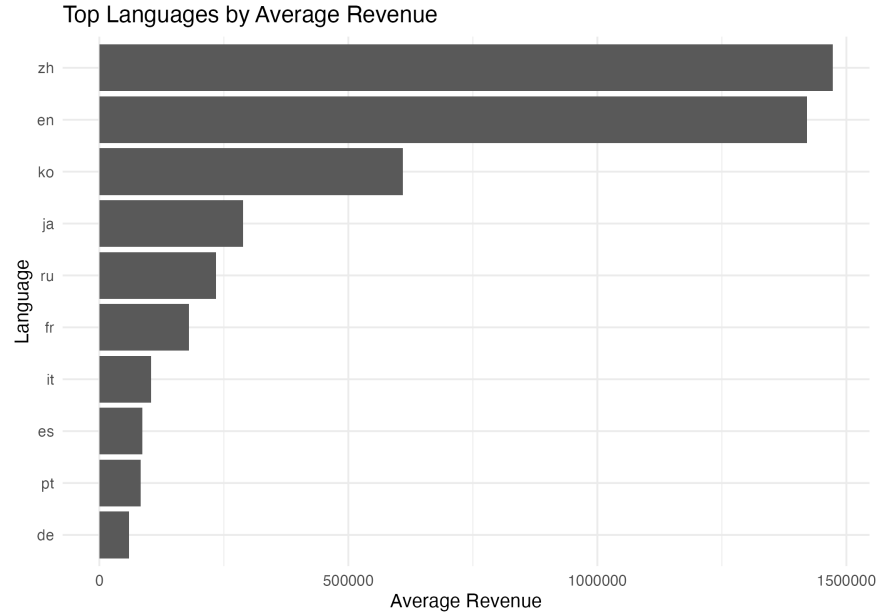
Genres were arranged based on revenue and rating because of the different importance of each factor for the content-acquisition team - revenue is important as the return, rating as reception.



Most Movies are distributed around a low to moderate ROI (0-5x), with a long tail past 40x – which implies that even an industry with a high average income might have been due to a few outstanding Movies.

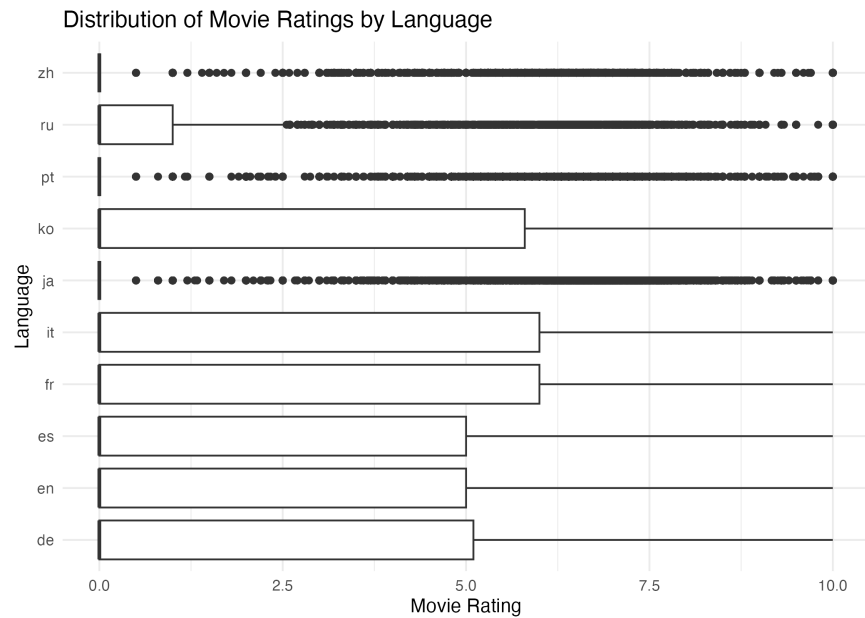


Influence of Language on Movie Success



Between the top 10 languages, it has been found that Chinese and English earn maximum average income, with Chinese earning slightly more than English. This could imply that there are other options for English-only approaches that have been overlooked.

Connection between language and ratings



Italian and French rank highest in median ratings among common languages, a pattern distinct from the revenue pattern, demonstrating that there is no necessary link between success as a language and success in terms of money or praise.

Evaluation - Cycle 2

Key Findings

- Adventure performs better than others on average revenue and rating.
- Drama has the largest number of movies but does not perform well on either of these two criteria, implying surplus in relation to reward.
- Chinese and English produce the highest average revenue, whereas Italian and French have the highest ratings.
- There is no significant correlation between revenue and rating – high revenue movies do not necessarily have the highest ratings.

Link to Success Criteria

These results meet the success criteria: a clear pattern emerged (Adventure and languages other than English both correlate with increased reception), and the correlation between revenue and ratings was shown to be largely independent. This provides a practical approach to decision making regarding content – emphasizing Adventure to generate revenue, and multilingual content for increased ratings.

Limitations

- “Genre” and “Language” are categorical variables that could misclassify subtle groups within the audience.
- There is no adjustment made for other variables, such as marketing budget or release date.

Next Steps

Future research may consider joint modeling of revenue and rating (such as through regression) in order to tease out the genre and language effects while controlling for the budget and year of release.

Future cycles would benefit from a separate indicator of success, independent of the revenue/rating thresholds.

Final Evaluation

Major Findings

- Chinese and English language films earn the most money.
- The genre that earns most money is adventure.
- Italian and French have the highest ratings among common languages.
- Rating and revenue are independent – the most revenue and highest rated films are different.

Implications for Film Production Industry

- The diversification to several languages can prove helpful in creating a good balance between the two aspects of earning and rating as well.
- The selection of Adventure as the genre for making the product would definitely prove helpful in achieving revenue.
- The use of Italian and French languages can prove helpful in increasing the rating.
- As high ratings cannot assure high earnings, the importance of quality and language cannot be ignored.

Study Limitations

- There are many missing values in some columns in the dataset, but the deletion of such observations wasn't performed in order not to reduce the number of cases, but that is the reason why there are several analyzes conducted on partially incomplete data.
- The results are drawn from one dataset (TMDb).
- There's no specific success variable in the source data – it was derived using the revenue/rating criteria. There are also certain observations with poor data quality characteristics instead of being real movies (unrealistic revenue in contrast with almost zero budgets and votes).

Conclusions

In both CRISP-DM Cycles, budget is revealed to be the most significant determinant of revenues, and ratings have little relationship with commercial success. Adventure is the best-performing genre in both categories, while Italian and French-language movies receive the most successful ratings even though there are fewer of them produced. It seems clear for the content acquisition team that adventure movies should be acquired for maximum revenue generation, while attention to other languages will improve ratings.

Methodological Notes

- The analysis was performed with the help of the CRISP-DM framework, implemented in two iterative rounds.
- The project relies on the ProjectTemplate framework for replicability.
- Round one analyzed global trends for the whole dataset; round two concentrated on genre and language.
- All data wrangling and analyses were performed in R with the help of dplyr.
- Visualizations were created with the help of ggplot2, apart from the correlation matrix that uses corrrplot.

References

- Bemthuis, R. H., Govers, R. R., & Asadi, A. (2025). A CRISP-DM-based methodology for assessing agent-based simulation models using process mining [Review of A CRISP-DM-based methodology for assessing agent-based simulation models using process mining]. *Journal of Simulation*, 1-22. <https://doi.org/10.1080/17477778.2025.2508245>
- Singh, P., & Asthana, S. (2025). Financial Sustainability in the Film Industry: Challenges, Strategies, and Future Directions. *International Journal on Science and Technology (IJSAT)* IJSAT25033691, 16(3). <https://www.ijSAT.org/papers/2025/3/3691.pdf>